

# NoteTrace

A safety net for AI scribe notes — it catches hallucinations and errors before they reach the patient record.

AI scribes save time, but they can invent a diagnosis, miswrite a dose, or put a procedure on the wrong side. NoteTrace compares the AI-written note against the actual consultation (the transcript, or any source you trust) and flags clinical detail:

 **verified** ·  **plausible** ·  **conflicting** ·  **needs review**

It understands clinical language, not just words — it knows "panadol" is paracetamol and "MI" is a heart attack — and it checks medication doses and procedure left/right, catching slips a simple text search would miss. When it can't confirm a finding, it points to the closest matching part of the source — so you know where to look, not just that something's missing. See [worked examples](#) of each.

CONSULTATION (SOURCE)

Dr: Any past history?

Pt: I had a **heart attack** last year, and I've got **diabetes**.

Dr: And current medications?

Pt: Just **Diabex 500 mg**, twice a day.

AI SCRIBE NOTE

PMHx: **MI**, **Type 2 diabetes**, **asthma**.

Meds: **Metformin hydrochloride 1g BD**.

FINDINGS

|              |                         |                                                         |
|--------------|-------------------------|---------------------------------------------------------|
| VERIFIED     | MI                      | matches "heart attack" in the consultation              |
| PLAUSIBLE    | Type 2 diabetes         | source only says "diabetes" — LLM adds more detail here |
| NEEDS REVIEW | asthma                  | nothing in the consultation supports it                 |
| CONFLICTING  | Metformin hydrochloride | dose 1 g ≠ 500 mg in the consultation                   |

And when a finding isn't supported, it points to the closest related line — not a silent miss:

AI SCRIBE NOTE

**Haematuria**  **needs review** — never stated in the consultation

NoteTrace can't confirm it — but instead of going silent, it points to the closest related line: ↓

CONSULTATION (SOURCE)

Dr: What brings you in today?

Pt: A cough and sore throat for a week.

Dr: Any fevers?

Pt: On and off, and I've felt really tired.

Dr: How's your appetite?

Pt: Not great, eating less than usual.

Pt: Oh — and my **urine** has looked a strange **dark colour** lately.

Dr: Any pain passing urine?

closest match · ~59%

# Beyond AI scribes

NoteTrace checks any LLM-generated clinical text against its source:

- **Ambient / AI scribe notes** — consultation transcript → note
- **Discharge & referral letters** — chart → generated letter
- **Note summarisation & handover** — long notes → short summary
- **Any clinical LLM output** you can pair with a source document

## Download

Latest builds for Linux and Windows (x86\_64):

[github.com/pisong314/notetrace/releases/latest](https://github.com/pisong314/notetrace/releases/latest)

| File                                                       | Platform                                   |
|------------------------------------------------------------|--------------------------------------------|
| <code>notetrace-&lt;version&gt;-linux-x86_64.tar.gz</code> | Linux glibc 2.28+ (RHEL 8 / Ubuntu 20.04+) |
| <code>notetrace-&lt;version&gt;-windows-x86_64.zip</code>  | Windows 10/11, Server 2019+                |

## Runs locally

**Standalone** — runs entirely on your own computer. Nothing is sent to the cloud; patient data never leaves your practice.

**Requirements:** x86\_64 CPU (no GPU), 256 MB RAM, 350 MB disk. Runs fully offline.

## Quickstart

Unzip what you downloaded, then send a `{summary, source}` pair on stdin:

```
echo '{"summary":"pt takes paracetamol 1g QID",
      "source":"patient takes panadol 500 mg four times a day"}' \
| ./notetrace-json
```

Run from the unzipped dist directory so `./data` is auto-discovered, or point at it explicitly with `export NOTETRACE_DATA_DIR=/path/to/dist/data` (or `--data-dir`).

Output (truncated):

```
{
  "items": [
    { "entity_text": "paracetamol",
      "provenance": { "level": "conflicting", "match_kind": "exact",
                     "source_text": "panadol", "confidence": 1.0,
                     "checks": [ { "attribute": "dosage", "status":
"conflicts",
                                "summary_value": "1g", "source_value":
"500 mg" } ] } }
```

```
],  
  "stats": { "total": 1, "verified": 0, "plausible": 0, "conflicting": 1,  
    "needs_review": 0 }  
}
```

For a long-running server, use `notetrace-server` — see [docs/rest.md](#). Full input/output schema is in the bundled `README.txt`.

## More

- [docs/examples.md](#) — what NoteTrace catches: faithful summaries, specification hallucinations, unsupported content, dosage mismatches, ...
- [docs/python.md](#) — Python API: load the engine, get a provenance report, inspect each item
- [docs/rest.md](#) — REST server and one-shot JSON CLI
- [docs/config.md](#) — runtime overrides: boilerplate filters, custom synonyms, semantic-type whitelist

## Known Limitations

- **Base vs salt drug names.** "Metformin" and "metformin hydrochloride" are separate concepts, so one may not match the other across note and source.
- **No negation or family-history detection yet.** "No chest pain" or "family history of diabetes" are treated as if the condition is present. Reliable handling needs more compute (GPU); this build is CPU-only for now.

## Reporting issues or coverage gaps

Bugs and feature requests → [Issues](#).

Please include the dist version, your OS, and a minimal repro (de-identified). The issue templates prompt for these.

**Coverage and tuning are ongoing.** If a clinical concept that should clearly be supported is being flagged as unsupported — or a hallucination is being labelled as verified — open an Issue with the input snippet, the expected provenance, and what you got. Your reports help calibrate the engine.

## Disclaimer

NoteTrace is a **review aid, not a medical device**. It flags content for a clinician to check; it does not approve, reject, or correct notes, and it is not a substitute for clinical judgement. A `verified` result means the source supports the statement — not that the statement is clinically correct. Always review the note against the full record before relying on it.

Maintained by [piyawoot.song@gmail.com](mailto:piyawoot.song@gmail.com). Questions and feedback welcome.